

Erdős Problem #786

EP-786

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $\epsilon > 0$. Is there some set $A \subset \mathbb{N}$ of density $> 1 - \epsilon$ such that $a_1 \cdots a_r = b_1 \cdots b_s$ with $a_i, b_j \in A$ can only hold when $r = s$?

Similarly, can one always find a set $A \subset \{1, \dots, N\}$ with this property of size $\geq (1 - o(1))N$?